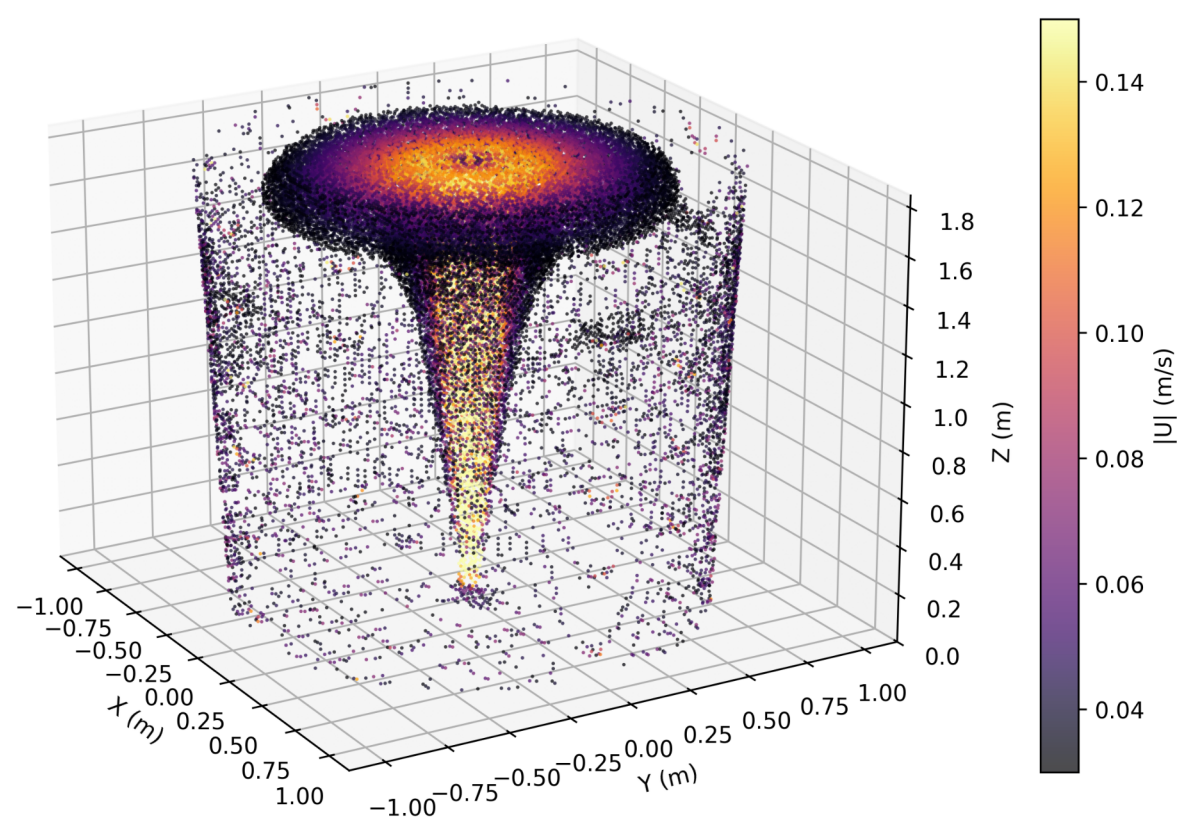


Plume core and velocity field visualization ($|U| > 0.03$ m/s, steel phase)

Center blowing

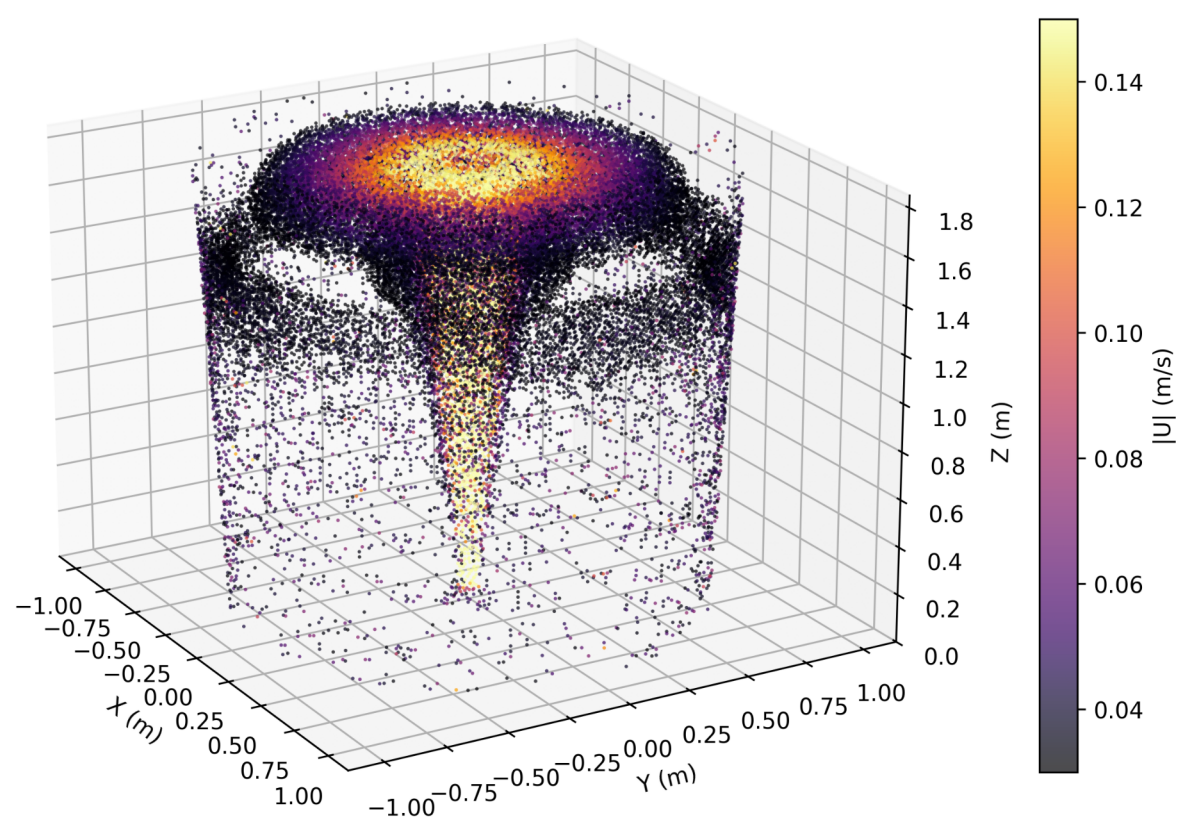
(a) Q=40 (Train)

Q=40 NL/min | Plume core ($|U| > 0.03$ m/s)



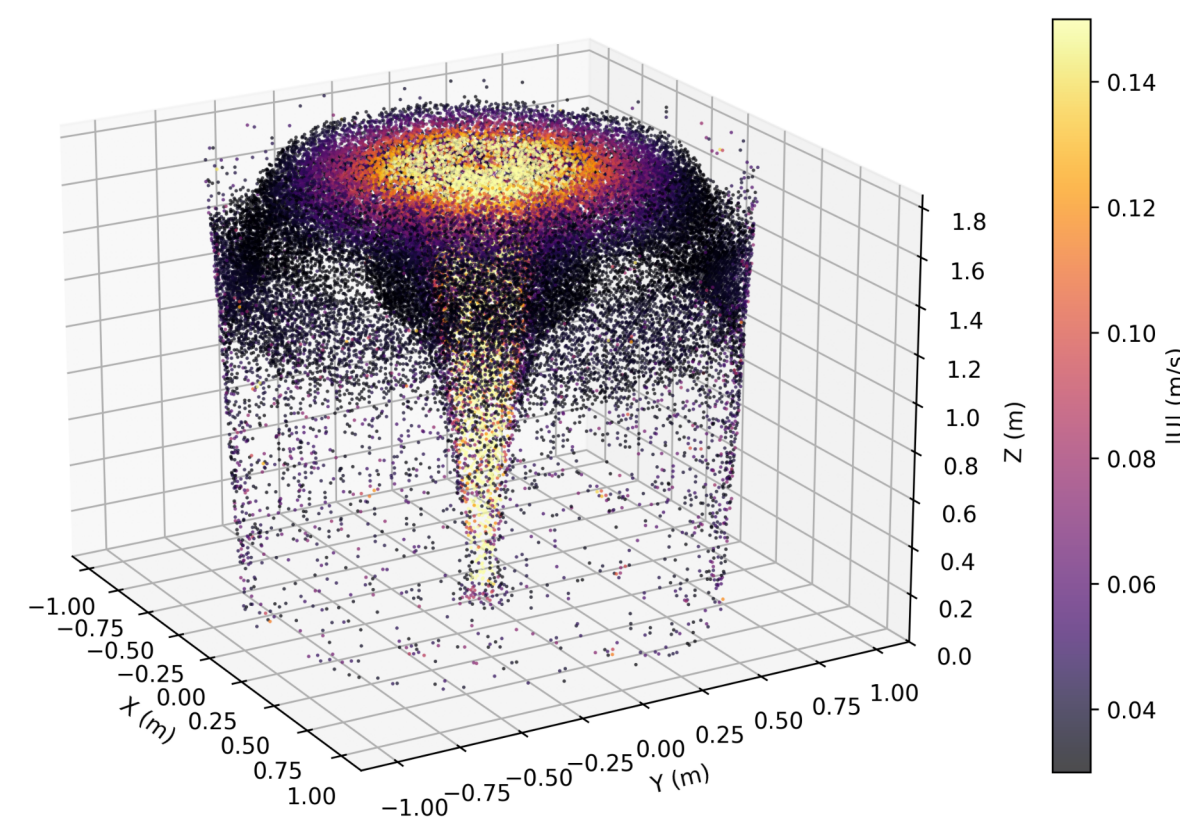
(b) Q=60 (Train)

Q=60 NL/min | Plume core ($|U| > 0.03$ m/s)



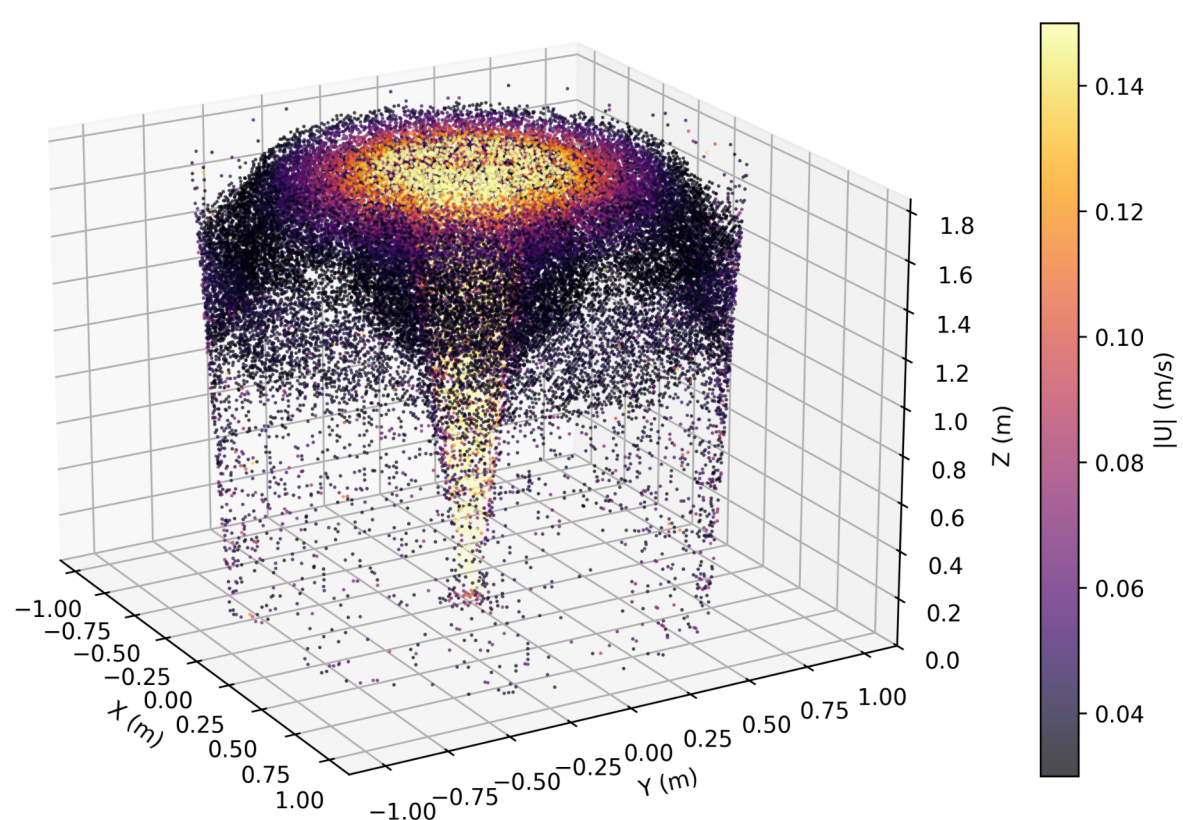
(c) Q=80 (Train)

Q=80 NL/min | Plume core ($|U| > 0.03$ m/s)



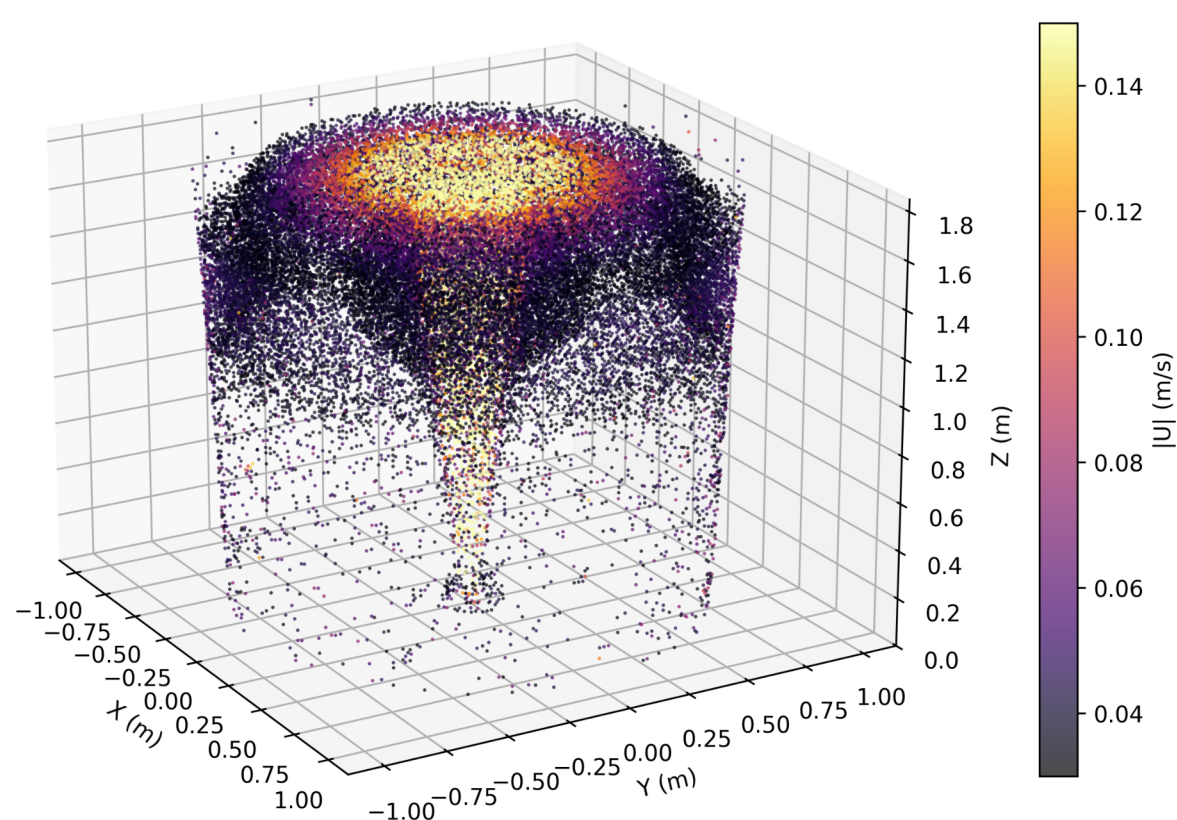
(d) Q=100 (Extrap.)

Q=100 NL/min | Plume core ($|U| > 0.03$ m/s)



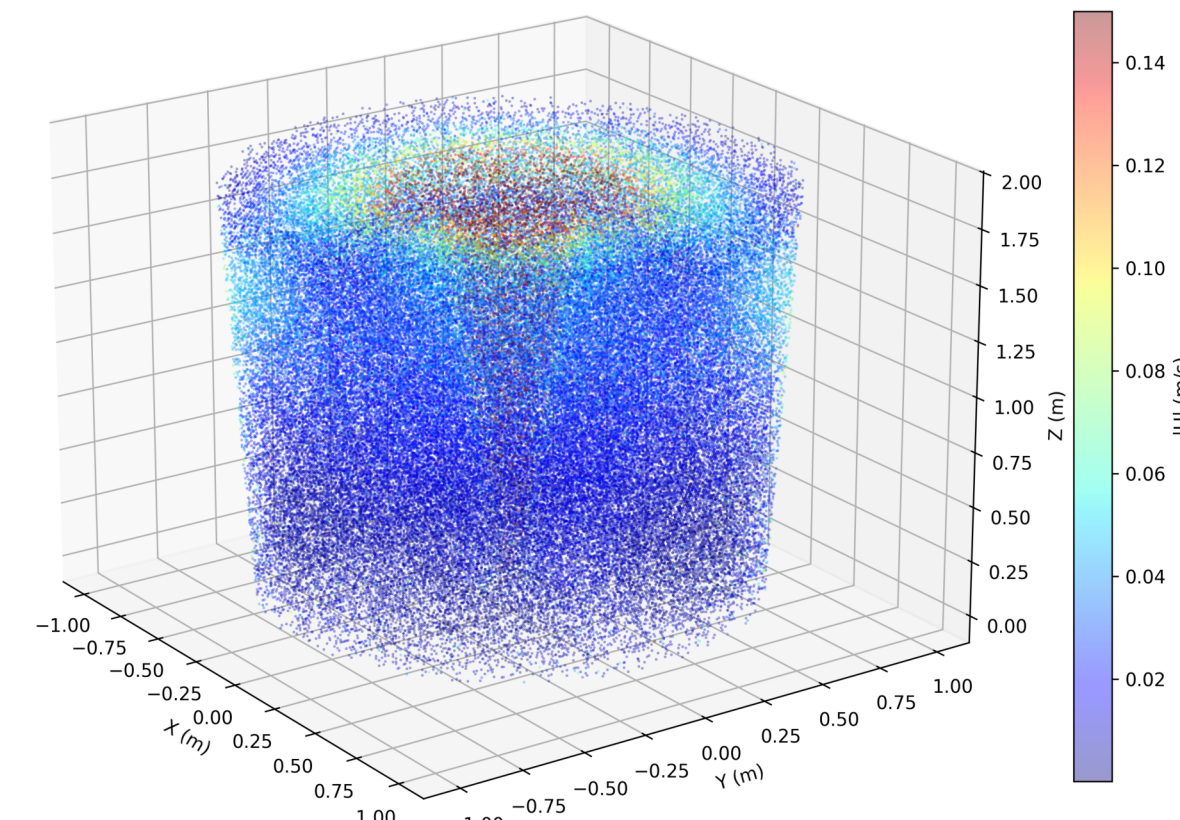
(e) Q=120 (Extrap.)

Q=120 NL/min | Plume core ($|U| > 0.03$ m/s)



(f) Q=120 (steel volume)

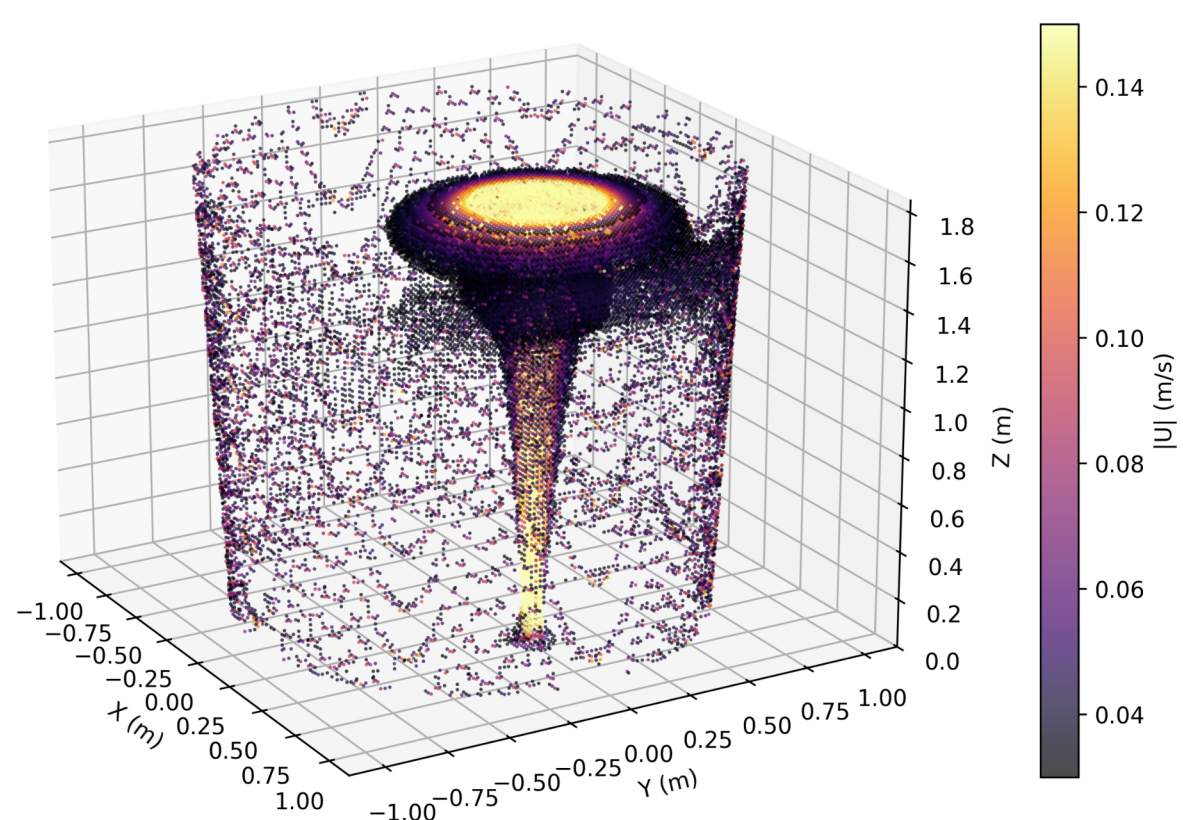
Q=120 NL/min | Steel phase velocity field



Center blowing

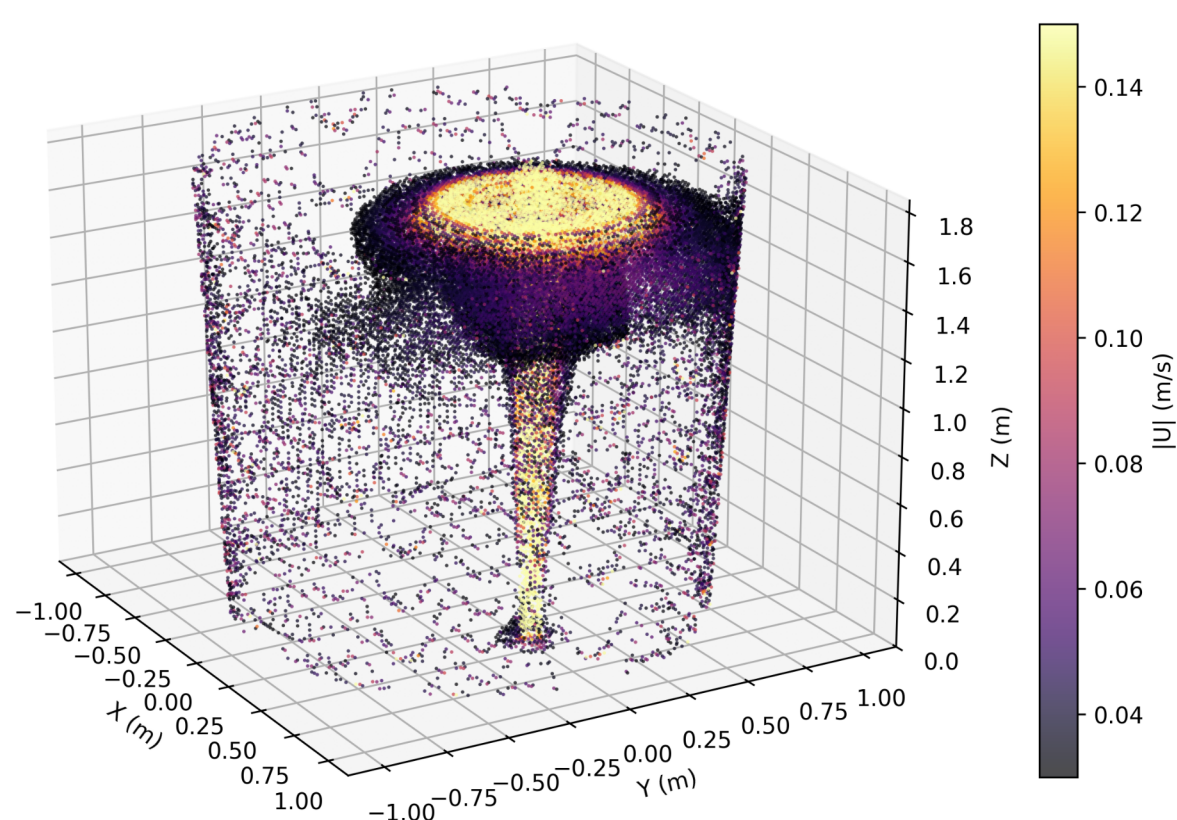
(g) Q=40

Q=Side Q40 NL/min | Plume core ($|U| > 0.03$ m/s)



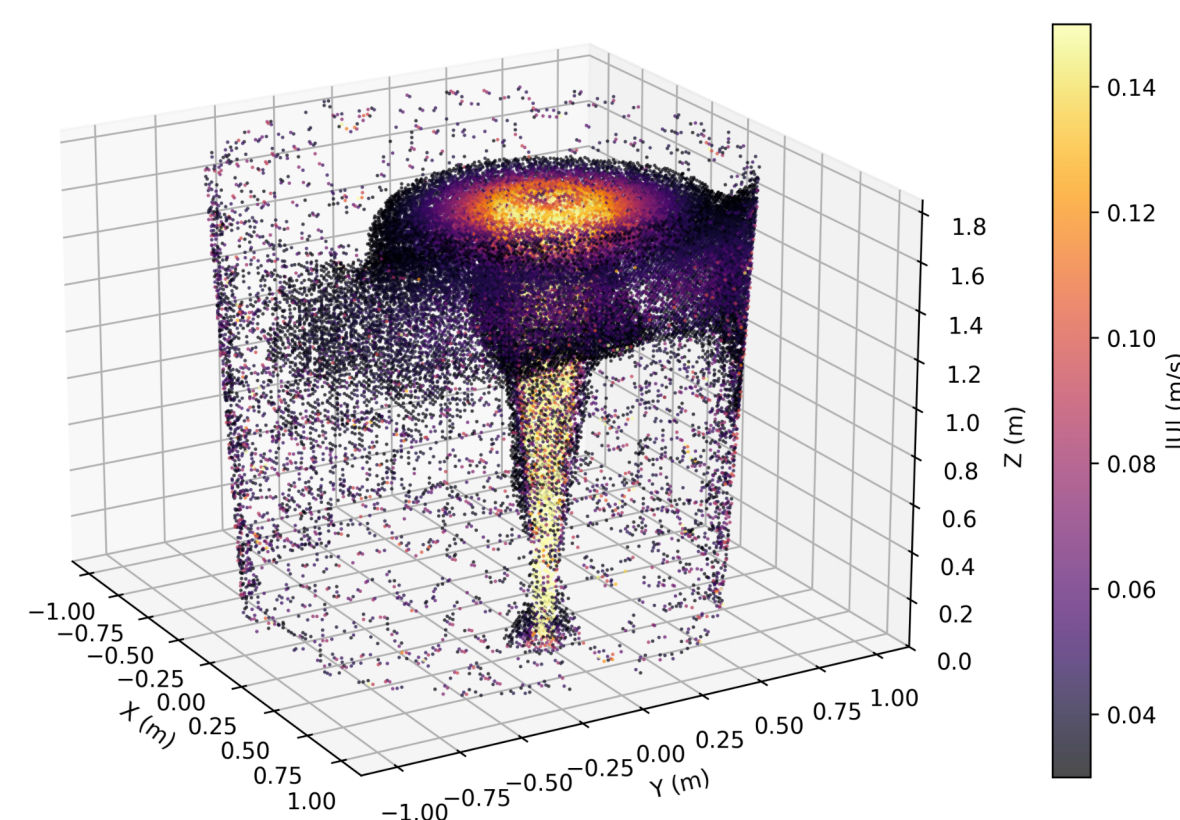
(h) Q=80

Q=Side Q80 NL/min | Plume core ($|U| > 0.03$ m/s)



(i) Q=120

Q=Side Q120 NL/min | Plume core ($|U| > 0.03$ m/s)



Eccentric blowing